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An improved artificial rabbits optimization for accurate and efficient infinite impulse response system identification

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ABSTRACT

Identifying models with Infinite Impulse Response (IIR) is crucial in signal processing and system identification. This paper addresses the challenges of IIR model identification by proposing an improved version of the Artificial Rabbits Optimization (ARO) algorithm called improved ARO (IARO). The IARO algorithm integrates an adaptive local search mechanism and an experience-based perturbed learning strategy as two key enhancements to improve the effectiveness of ARO. These additions aim to address the loss of accuracy during iterations and improve the algorithm's ability to exploit promising areas. Four benchmark examples of different IIR plants are considered, and the performance of the proposed IARO algorithm is compared to existing competitive methods. The results consistently demonstrate that IARO outperforms ARO in accuracy and convergence for system identification across all orders of IIR systems. Visual analysis, convergence curves, coefficient comparison, and statistical metrics comparison all validate the superiority of the IARO algorithm. Additionally, the Wilcoxon signed-rank test results provide further statistical evidence supporting the superior performance of IARO. The comprehensive analysis showcases the efficacy and effectiveness of the IARO algorithm in accurately identifying IIR systems. This work represents a significant advancement in IIR system identification, offering a superior methodology for accurate and efficient system modeling.

1. Introduction

Identifying models with infinite impulse response (IIR) is crucial in signal processing and system identification. IIR models are commonly used to represent systems that have long-term memory, such as filters and control systems. They offer a concise representation and efficient computational solutions [1–5]. However, the identification of IIR models can be challenging due to their inherent characteristics, necessitating specialized techniques like maximum likelihood or subspace identification [6,7]. Accurate identification of IIR models is vital for the design and optimization of systems relying on them, such as digital signal processing and control systems [8–11]. Moreover, it enhances the understanding and analysis of complex systems, facilitating more effective and efficient decision-making.

In the literature, various metaheuristic algorithms have been proposed to tackle the problem of IIR system identification [12–16] due to greater ability of such algorithms to achieve near-global optimum solutions both linear and non-linear plants [17–22]. For instance, researchers have explored the application of gravitational search algorithm, inclined planes system optimization, genetic algorithm and particle swarm optimization for the design and optimization of digital IIR filters [23]. The performance and efficiency of these methods have been evaluated using mean squared error (MSE), demonstrating the successful achievement of accurate results. In a recent study [24], an enhanced version of the golden jackal optimization method was proposed, incorporating the elite opposition-based learning technique and the simplex technique, to address system identification problems, demonstrating superior efficiency, accuracy, convergence rate, precision, control parameters, and fitness value compared to other optimization algorithms. In another study [25], a modified version of the whale optimization algorithm was proposed to address the limitations of the original algorithm. By integrating opposition-based learning, exponentially decreasing function, and elite-guided Cauchy mutation, the proposed algorithm enhances exploration performance, avoids local optima, and improves solution quality and convergence rate, as demonstrated through experiments on various benchmark functions and IIR system identification tasks, outperforming seven commonly used algorithms. A different modified whale optimization algorithm has also been reported in another study for IIR identification [26]. In [27], an average differential evolution with local search was introduced for identifying optimal coefficients of unknown IIR systems.

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It is feasible to list more studies in this regard as several other metaheuristic approaches have been reported for the IIR system identification problem, such as cuckoo search algorithm [28] and it modified version [29], hybrid chimp-cuckoo algorithm [30], a sigmoid attractiveness based improved firefly algorithm [31], an enhanced slime mould algorithm [32], teacher learner-based optimization [33], grey wolf optimizer with ranking-based mutation operator [34], modified-interior search algorithm with Lèvy flight [35], firefly algorithm [36], hybrid cellular particle swarm optimization and differential evolution [37], seeker optimization algorithm [38], artificial bee colony algorithm [39], a modified version of particle swarm optimization and artificial bee colony algorithms [40,41], quantum particle swarm optimization and artificial bee colony algorithm [42]. These examples demonstrate the feasibility of employing various metaheuristic techniques to address the challenges of IIR system identification.

In the ever-continued scape of signal processing and system identification, the quest for efficient and accurate methodologies continues to drive researchers towards new innovations [43]. With an insatiable appetite for cutting-edge solutions, researchers and practitioners seek novel approaches to tackle the challenges inherent in IIR system identification. Traditional methods, while effective to a certain extent, are often plagued by limitations such as slow convergence, sensitivity to initial conditions, and suboptimal performance. Recognizing these drawbacks, the research community has been eagerly awaiting new methods that can transcend the boundaries of existing techniques and unlock new possibilities. In this pursuit, the development of intelligent algorithms has played a pivotal role in unraveling complex patterns and uncovering hidden insights within intricate systems [44–50]. In line with this motivation, this work proposes a new optimizer by improving the performance of one of the recent metaheuristic algorithms known as artificial rabbits optimization (ARO) algorithm [51].

The ARO algorithm offers simplicity in implementation and fast convergence to good solutions by identifying the best participants within the search region. However, the algorithm suffers from significant flaws that hinder its ability to handle complex and multimodal problems while striking a balance between exploration and exploitation. One major issue is that the rabbits in ARO perform the iterative process by randomly selecting a burrow, which can accelerate convergence but may degrade solution diversity and trap the rabbits in local optima. Furthermore, ARO lacks a guidance strategy to navigate towards promising regions during the iterative search, leading to potential degradation in the quality of the final solution. Overcoming this challenge of ensuring the new population reaches prospective areas presents an excellent opportunity for improvement. In essence, there is a need to enhance the effectiveness of the traditional ARO algorithm. To address these shortcomings, this paper proposes an improved version called IARO, which integrates an adaptive local search (ALS) mechanism and an experience-based perturbed learning (EPL) strategy.

Four benchmark examples of different IIR plants, along with their corresponding same and reduced order IIR filter models. The examples range from second order systems to fifth order systems, and each is evaluated using both the original form of the ARO algorithm [51] and the proposed IARO algorithm to identify the coefficients of the reduced and same orders digital IIR filters in order to estimate the unknown plants. The objective of this article is to showcase the remarkable capabilities of IARO in comparison to existing approaches.

By conducting a comprehensive empirical evaluation and performance analysis, we aim to demonstrate the superior effectiveness and efficiency of IARO in identifying IIR systems. In this context, coevolution particle swarm optimization with multi-strategy [52], differential evolution [52], differential evolution with a hybrid mutation operator and self-adapting control parameters [52], and opposition-based hybrid coral reefs optimization algorithm [52] are considered as different existing competitive methods for the second-order system modeled with reduced and same order IIR filters. For the third-order system modeled with reduced and same order IIR filters, different methods, including cat swarm optimization [53], firefly algorithm [54], hybrid particle swarm optimization and gravitational search algorithm [54], and selfish herd optimization algorithm based on chaotic strategy [54] are considered as different existing competitive methods. Similarly, for the fourth-order system modeled with reduced and same order IIR filters, bat algorithm [55], Lévy flight trajectory-based whale optimization algorithm [55], grey wolf optimizer with ranking-based mutation operator [55], and dynamic opposite learning enhanced artificial ecosystem optimizer [55] are considered as reported methods. Lastly, for the fifth-order system modeled with reduced and same order IIR filters, genetic algorithm [23], gravitational search algorithm [23], inclined planes system optimization [23], and particle swarm optimization [23] are considered.

The results consistently indicate that the IARO algorithm surpasses the conventional ARO algorithm in terms of accuracy and convergence for system identification. Regardless of the order of IIR systems, IARO consistently achieves lower MSE values, demonstrating its superior ability to minimize the discrepancy between estimated and actual system outputs. Visual analysis of scatter plots reveals a consistent pattern across all orders of IIR systems, with IARO generating data points predominantly lower than those obtained through ARO. This suggests that the IARO algorithm effectively finds optimal solutions that closely match the actual system behavior. Convergence curves further confirm this trend, as IARO consistently converges to lower MSE values compared to ARO, showcasing its superior ability to explore the search space efficiently and identify optimal solutions. The analysis of coefficient comparison supports the effectiveness of the IARO algorithm in optimizing system parameters. In all cases, the coefficients obtained through IARO exhibit closer values to the real values compared to ARO, indicating its superior ability to fine-tune system parameters and improve system identification accuracy. Furthermore, statistical metrics comparison validates the superiority of the IARO algorithm. Across all orders of IIR systems, IARO consistently achieves the best values for metrics such as best, worst, mean, and standard deviation. The best values obtained through IARO are significantly closer to the real values than those obtained through ARO and other alternative methods, highlighting the algorithm's consistent generation of solutions with superior performance and accuracy. Additionally, the Wilcoxon signed-rank test results further supports the superior performance of IARO. In summary, the comprehensive analysis of the IARO algorithm for IIR system identification demonstrates its efficacy and superiority over the conventional ARO algorithm and other existing methods as it consistently achieves lower MSE values, better convergence, optimized system coefficients, and superior statistical metrics. These findings underscore IARO's ability to effectively explore the search space, identify optimal solutions, and accurately model underlying IIR systems. Therefore, IARO algorithm represents a significant advancement in the pursuit of superior effectiveness in IIR system identification.

The contributions and novelty of this work lie in the introduction of the IARO algorithm, its enhancements over ARO, the integration of novel strategies (ALS and EPL), and its superior performance demonstrated through comprehensive empirical evaluation and benchmarking against existing optimization methods in the context of IIR system identification. In line with the discussion so far, the novelty of this work and the contributions can briefly be highlighted as follows.

(1) This work introduces the IARO algorithm, addressing the limitations of existing optimization methods for IIR system identification.
(2) IARO introduces significant enhancements to the traditional ARO algorithm, making it more effective in solving complex problems, which is a novel contribution to the field of system identification.
(3) IARO integrates an ALS mechanism and an EPL strategy to enhance its performance, offering a unique approach to optimization. The integration of those mechanisms within IARO represents a novel approach that improves its ability to navigate complex search spaces.

(4) The study conducts a comprehensive empirical evaluation, comparing IARO with a wide range of competitive optimization algorithms, including particle swarm optimization, genetic algorithms, and various hybrid methods. The extensive benchmarking of IARO against a diverse set of competitive algorithms in IIR system identification is a novel aspect, demonstrating its effectiveness and superiority across various scenarios.
(5) Results consistently show that IARO outperforms ARO and other existing methods in terms of accuracy, convergence, and the ability to identify optimal solutions for IIR system identification. IARO consistently achieves lower MSE values, better convergence, and optimized system coefficients, showcasing its consistent and superior performance compared to existing methods.
(6) The validation of statistical metrics further highlights the novelty of IARO in consistently generating superior solutions for system identification.

The remaining sections of the paper are organized as follows: Section 2 provides a brief explanation of the ARO algorithm and its mathematical structure. Section 3 explains the proposed IARO algorithm. Section 4 presents the details regarding IIR system identification and the design procedure using digital IIR filtering systems to accurately represent unknown plants. The subsequent section presents simulation results obtained for different benchmark examples and discusses the significance of the IARO-based design method compared to other state-of-the-art existing methods. Finally, the paper concludes in Section 6.

2. The conception of artificial rabbits optimization

The artificial rabbits optimization (ARO) algorithm is a recent and potent metaheuristic technique inspired by the survival natures of rabbits [51], such as random hiding and detour foraging. The foraging mode of the ARO is modeled mathematically to perform the iterative searches. In this sense, A rabbit attempts to eat the plants and grasses adjacent to other rabbits' burrows. This action might aid in fooling predators and defending the rabbit burrow from attacks. In other words, rabbits ignore the food that is readily available nearby and instead look for food in faraway locations. The ARO algorithm uses the number of rabbits as the swarm population. Each rabbit has an eating zone with some grass and plants in addition to a d burrows. Each rabbit randomly visits the burrows of other rabbits in search of food. Each rabbit tends to update its current location with respect to the randomly chosen individual and contributes a disruption throughout this stage. This foraging action can be represented mathematically as follows [51]:

$$z'_i(t+1) = z'_j(t) + \ldots z'_i(t) - z'_j(t) + \text{round}\ 0.5:\ 0.05 + g_1 \quad :n_1; i; j = 1; 2 \ldots \ldots; M \text{ and } i \neq j \tag{1}$$

$$= E:c \tag{2}$$

$$E = \quad e - e^{\left(\frac{1-t}{T}\right)^2} : \sin(2\ g_2) \tag{3}$$

$$c(k) = \begin{cases} 1 & \text{if } k == h(u) \\ 0 & \text{else} \end{cases} ; k = 1; \ldots; d \&u = 1; 2; \ldots; \text{äg}_3:\text{då} \tag{4}$$

$$h = \text{randperm}(d) \tag{5}$$

$$n_1 \sim N(0; 1) \tag{6}$$

where $z'_i(t)$; $'_i(t+1)$; M; T; d; round; randperm, and E define the current position of candidate of ith candidate at time t, ith updated candidate at time t + 1, population size of rabbits, the total size of iterations, the problem dimension, the rounding to the nearest integer value, random permutation function ranged from 1 to problem dimension, and running length during foraging, respectively. Here n_1 denotes normal distribution function, and $g_1; g_2; g_3$ define uniform random numbers inside the interval [0, 1]. ARO is assisted by the perturbation in Eq. (6) to avoid local maxima/minima and carry out a comprehensive search. The running distance (E) helps in conducting longer distances during the early iterations and shorter distances in the later iterations to perform the exploration and exploitation modes, respectively. Here represents a mathematical operator that simulates the locomotion of rabbits, and c is a vector that is used in the search process to select a number of individuals. Therefore, this foraging stage boosts the ARO algorithm's exploration and global search capabilities. In the exploitation mode, rabbits randomly hide to avoid being caught by predators. In close proximity to its burrows, the rabbit creates a few burrows. During each iteration of ARO, a rabbit consistently creates b burrows throughout each dimension and it chooses a burrow at random and hides there to fool the predators.

The ith rabbit with jth burrow can be formulated as follows [51]:

$$BU'_{ij}(t) = z'_i(t) + H:h:z'_i(t); i = 1; 2 \ldots \ldots; M \text{ and } j = 1; 2 \ldots \ldots; d \tag{7}$$

$$H = \frac{1 - t + T}{T} g_4 \tag{8}$$

$$n_2 \sim N(0; 1) \tag{9}$$

$$h(k) = \begin{cases} 1 & \text{if } k == j \\ 0 & \text{else} \end{cases} ; k = 1; \ldots; d \tag{10}$$

where H and d denote the hiding function, and formed burrows within the area of the rabbit, respectively. Primarily, a bunny's big area is where holes are generated. The size of this neighborhood shrinks as the amount of iterations rises. the random hiding mode is offered as follows [51].

$$'_i(t+1) = z'_i(t) + \ldots g_4:BU_{ir}(t) - z'_i(t) \quad ; i = 1; 2; \ldots \ldots M \tag{11}$$

$$h_r(k) = \begin{cases} 1 & \text{if } k == g_5:d \\ 0 & \text{else} \end{cases} ; k = 1; \ldots; d \tag{12}$$

$$BU_{ir}(t) = z'_i(t) + H:h_r:z'_i(t) \tag{13}$$

where $BU_{ir}(t)$ denotes the chosen burrow by the rabbit through the hiding mode, g_4 and g_5 define random numbers inside the interval [0, 1]. The ith rabbit's position is updated after either a detour foraging mode or a random hiding process as follows [51].

$$z'_s(t+1) = \begin{cases} z'_s(t) & f(z'_s(t)) \ f \ f('_s(t+1)) \\ '_s(t+1) & f(z'_s(t)) > f('_s(t+1)) \end{cases} \tag{14}$$

If the sth rabbit's candidate fitness is better than the position's present fitness, the bunny leaves its current location and stays in the candidate location determined by either Eq. (1) or Eq. (11). The energy of rabbits is shrinking with the progress of iteration, which helps in the transition from explorative mode to exploitative mode, which is formulated as follows [51]:

$$EA(t) = 4 \quad 1 - \frac{t}{T} \quad \ln \quad \frac{1}{} \tag{15}$$

where defines a random number. When $EA(t) > 1$, the algorithm searches globally for the solution (exploration); when $EA(t) \ f \ 1$, the algorithm searches locally for the solution (exploitation). The pseudo-code framework of the original ARO is presented in Fig. 1.

3. The proposed IARO algorithm

ARO can obtain the best participants in the search region with the advantages of its simplicity in implementation and the fast convergence to good solutions by estimating its development. The algorithm still has significant flaws, though, that could lead to signation dilemmas and prevent it from finding a balance between exploitative and exploratory patterns when it comes to handling multimodal and complex issues. Firstly, the rabbits perform the iterative process by choosing a burrow at random; although this scheme can accelerate the convergence pattern, the solution diversity may degrade, leading to the rabbits getting stuck in the local optimal solution. Firstly, no guidance strategy is

1. Initialize a set of rabbits randomly
2. Evaluate the fitness of each rabbit and determine the best one (z_{best})
3. **While** the stopping criterion not met **do**
4. **for** $i = 1 : M$
5. Compute the energy of rabbit (EA) by Eq. (15)
6. **if** EA > 1
7. Select a rabbit randomly from the population
8. Obtain ρ by Eqs. (2)-(6)
9. Carry out the detour foraging phase by Eq. (1)
10. Evaluate the fitness of the rabbit
11. Update the rabbit' position by Eq. (14)
12. **else**
13. Create d burrows and elicit one of them randomly as hiding position by Eq. (13)
14. Conduct random hiding by Eq. (11)
15. Evaluate the fitness of the rabbit
16. Update the rabbit' position by Eq. (14)
17. **End if**
18. Update the best so far solution (z_{best})
19. **End for**
20. **End while**
21. **Output:** Return z_{best}

Fig. 1. The framework of the ARO.

employed in the ARO to approach the promising regions during the iterative search, which may deteriorate the quality of the final solution. Therefore, figuring out how to make sure that the new population can get to the prospective area is a good challenge. In other words, there are opportunities to improve the effectiveness of the traditional ARO. Hence, an improved ARO, named IARO, based on adaptive local search (ALS) mechanism, and experience-based perturbed learning (EPL) strategy are proposed in this paper.

3.1. The prospective of EPL strategy

In the IARO algorithm, the rabbits are behaved by following another rabbit from the population during the exploration phase. This update scheme can lead to a pattern of invasive diversification. Therefore, the EPL is embedded to discover more promising regions inside the feasible search space, thus boosting the exploration search. In particular, the EPL starts by calculating the mean ($^{it}_{mean}$) and the deviation ($^{it}_{dev}$) of any solution chosen randomly with respect to the best so far solution ($_{best}$). Afterwards, the new solution is updated according the $^{it}_{mean}$ and $^{it}_{dev}$ along with perturbed term. Therefore, the updated solution can be expressed as follows.

$$^{it}_{mean} = (z_{best} + z^{it}_{l})/2 \tag{16}$$

$$^{it}_{dev} = \mathrm{abs}(z_{best} - z^{it}_{l}) \tag{17}$$

$$^{it}_{C} = {}^{it}_{mean} + \mathrm{rand}_1 : {}^{it}_{dev} \tag{18}$$

$$z^{it}_{new} = {}^{it}_{C} + \mathrm{rand}_2 : (z_{best} - {}^{it}_{C}) + 0{:}95^{it} : (\mathrm{rand}_3 - 0{:}5) : \mathrm{abs}(z_{max;j} - z_{min;j}); z_{max;j} = \max_j \; z^{it}_i \; ; z_{min;j} = \min_j \; z^{it}_i \quad \forall i \tag{19}$$

where $^{it}_{l}$ stand for any arbitrary solution chosen at random, rand_1; rand_2, and rand_3 define three random numbers elicited according to the uniform distribution inside the interval $[0; 1]$. Here, the third term of Eq. (19) is to perform perturbed solution within the dynamic bounders ($_{max}$ and $_{min}$).

3.2. The prospective of ALS strategy

To mitigate the loss of accuracy during the iterative process and enhance the exploitation trend towards the promising space, an adaptive local search (ALS) strategy is purported as a guidance scheme based on shared information among the elite group within the rabbits along with their best individual (z_{best}) and worst individual (z_{worst}). To be specific, the process by determining the elite group according to the fitness function as well as identifying the worst and best individuals within this grope (z^P and z^W). Afterwards, three types of movements are adopted to perform the updating step, which are pushing z^W in the direction of z^P, or pushing z^W towards z_{best}, and pushing z^W in the direction of the average of z^P and z_{best}. These movements are carried out in sequential manner and it stops after a certain one reaches better fitness. The updating phase of this strategy can be expressed as follows:

$$z^{it+1}_{l} = \begin{cases} z^{ALS}_1 = 2 \times r_1 \times \left(z^P - z^W \right) + z^W & \text{if } f(z^{ALS}_1) \; f \; f(z^{it}_l) \\ z^{ALS}_2 = 2 \times r_2 \times \left(z_{best} - z^W \right) + z^W & \text{else if } f(z^{ALS}_2) \; f \; f(z^{it}_l) \\ z^{ALS}_3 = 2 \times r_3 \times \left(z^P + z_{best} \right) /2 - z^W + z^W & \text{otherwise} \end{cases} \tag{20}$$

where z^{it}_l and z^{it+1}_l stand for the present and updated solutions of the ith solution inside the elite class. The framework of the proposed IARO is represented as in Fig. 2.

4. Problem definition of IIR system identification and recommended approach

System identification involves mathematically representing an unknown system based on input and output data [56]. The goal is to minimize an error function using an optimization algorithm, which finds the best model that closely matches the actual plant's output. IIR models offer the advantage of using fewer parameters while meeting performance requirements and providing accurate representations of physical plants for real-world applications [57]. Fig. 3 illustrates the implementation of the IARO algorithm for the IIR system identification problem.

In here, $y(k)$ and $d(k)$ respectively represent the output of the IIR filter and the unknown plant. On the other hand, $x(k)$ stands for the applied input signal whereas m and n are respectively the coefficients of the numerator and denominator. From the information provided in